TRI NHU DO

Curriculum Vitae

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Academic Experience

Research Interests

I am interested in both basic and applied research of wireless communications. In particular, the main focuses of my research are:

- Multiple-Input Multiple-Output (MIMO) Communications
- Applications of Machine Learning to Wireless Communications
- Power-Domain Non-Orthogonal Multiple Access (NOMA) Techniques
- Radio-Frequency Energy Harvesting (RF-EH) Techniques
- Physical Layer Security (PHY-Security)
- Cognitive Radio Networks
- Cooperative Communication Systems

Research Experience

Oct. 2019 to present — **Collaborative Researcher**, *Hongik University*, South Korea.
Department of Software and Communications Engineering
Ubiquitous Networks & Convergence Laboratory (UNC Lab)
Supervisor: *Prof. Beongku An (link)*

Mar. 2019 to Sep. 2019 — **Postdoctoral Research Associate**, *The Univeristy of Texas at Dallas*, TX, USA.
Department of Computer Science
Supervisor: *Prof. Zygmunt J. Haas (link)*

Our research focused on the theoretical performance of multiple-input multiple-output (MIMO) systems, in particular, the re-configurable distributed MIMO systems.

Mar. 2013 to Aug. 2018 — **Graduate Research Assistant**, *Hongik University*, South Korea.
Department of Software and Communications Engineering
Ubiquitous Networks & Convergence Laboratory (UNC Lab)
Supervisors: *Prof. Beongku An (link) and Prof. Daniel B. da Costa (link)*

I had participated as the role of Graduate Research Assistant in the following research projects (Certificate of Research Project Participation):

- From Dec. 01, 2014 to Nov. 03, 2015 "*Network Coding based Cooperative Transmission and Cooperative Routing in Mobile Cognitive Wireless Ad-hoc Networks*." This research project was supported by the Basic Research Program in Science and Engineering funded by the Ministry of Science, ICT and Future Planning (Grant No. 2013075605).
- From Dec. 01, 2015 to Jun. 30, 2016 : "*Used Spice Protocol, Mobile Virtual Desktop Infrastructure (VDI) Client Development (Friendly User Interface)*." This research project was supported by the ICT/SW Creative Research Program funded by the Information and Communication Technology Promotion Center (Grant No. IITP-2015-R2212-15-0026).
- From Oct. 01, 2016 to Jun. 30, 2018 : "*IoT based Mobile Smart Solar Power Generation Management System*." This research project was supported by the Leading Human Resource Training Program of Regional Neo-industry funded by the National Research Foundation of Korea (NRF) (Grant No. NRF-2016H1D5A1910577).

- From Nov. 01, 2016 to Jun. 30, 2018 : "*Development of Energy Efficiency and Robust Security-based Cooperative Relay Transmissions and Routing Protocols for Future Mobile Wireless Networks*." This research project was supported by the Basic Research Program in Science and Engineering funded by the National Research Foundation of Korea (NRF) (Grant No. 2016R1D1A1B03934898).

Teaching Experience

Mar. 2017 to Aug. 2018 **Teaching Associate**, *Hongik University*, South Korea.
Department of Software and Communications Engineering
Teaching Certificate (link)

Teaching undergraduate courses:

[012301] Computer Engineering: Introduction and Practice *Sping–2017/2018*

[704410] Computer Architecture *Fall–2017*

Education

Mar. 2015 to Aug. 2018 **Doctor of Philosophy**, *Hongik University*, South Korea.
Electronics and Computer Engineering

Thesis: *"Cooperative Transmission Schemes for Energy and Spectral Efficient Wireless Networks"*

Advisor: *Prof. Beongku An (link)*

Mar. 2013 to Feb. 2015 **Master of Science**, *Hongik University*, South Korea.
Electronics and Computer Engineering

Thesis: *"Centralized Cooperative Spectrum Sensing Schemes for Cognitive Radio Networks"*

Advisor: *Prof. Beongku An (link)*

Sep. 2007 to Feb. 2012 **Bachelor of Engineering**, *Posts and Telecommunications Institute of Technology*, Vietnam.
Electronics and Telecommunication Engineering

Thesis: *"A Study on Multiple-Input Multiple-Output (MIMO) Technique in Evolved High Speed Packet Access (HSPA+) Networks"*

Advisor: *Prof. Vo Nguyen Quoc Bao (link)*

Publications

In Preparation

- **Tri Nhu Do**, Tung Tien Nguyen, Toan Van Nguyen, Daniel Benevides da Costa, and Zygmunt J. Haas, "Deep Neural Network-enabled Dynamic Distributed Relaying Schemes in NOMA Systems", 2020 *(intended to submit to IEEE Wireless Communications Letters)*

- Kyusung Shim, Toan Van Nguyen, **Tri Nhu Do**, Daniel Benevides da Costa, and Beongku An, "Improving PHY-Security of RF-EH and Full-Duplex-based MISO-NOMA Systems using TAS," 2020.

2020

- Toan Van Nguyen, **Tri Nhu Do**, Daniel Benevides da Costa, Vo Nguyen Quoc Bao, and Beongku An, "On the Performance of Multihop Cognitive Wireless Powered D2D Communications in WSNs," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 3, pp. 2684-2699, Mar. 2020

- **Tri Nhu Do** and Zygmunt J. Haas, "On the Design of RD-MIMO: Spatial Multiplexing versus Opportunistic Transmission Schemes," *IEEE Access*, vol. 8, pp. 22733-22747, Jan. 2020.

2019

- **Tri Nhu Do**, Van-Dinh Nguyen, Oh-Soon Shin and Beongku An, "Simultaneous Uplink and Downlink Transmissions for Wireless Powered Communication Networks," *IEEE Communications Letters*, vol. 23, no. 2, pp. 374-377, Feb. 2019.

- Toan-Van Nguyen, **Tri Nhu Do**, Vo Nguyen Quoc Bao, Daniel Benevides da Costa, and Beongku An, "Performance Analysis of Multihop Cognitive WPCNs with Imperfect CSI," in *Proc. IEEE Global Communications Conference (GLOBECOM 2019)*, Waikoloa, Hawaii, USA, Dec. 2019.

- Toan-Van Nguyen, Van-Dinh Nguyen, **Tri Nhu Do**, Daniel Benevides da Costa, and Beongku An, "Spectral Efficiency Maximization for Multiuser MISO-NOMA Downlink Systems with SWIPT," in *Proc. IEEE Global Communications Conference (GLOBECOM 2019)*, Waikoloa, Hawaii, USA, Dec. 2019.

2018

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Improving the Performance of Cell-Edge Users in NOMA Systems Using Cooperative Relaying," *IEEE Transactions on Communications*, vol. 66, no. 5, pp. 1883–1901, May 2018.

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Improving the performance of cell-edge users in MISO-NOMA systems using TAS and SWIPT-based cooperative transmissions," *IEEE Transactions on Green Communications and Networking*, vol. 2, no. 1, pp. 49–62, Mar. 2018.

- Nguyen Toan Van, **Tri Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "Performance analysis of wireless energy harvesting multihop cluster-based networks over Nakgami-m fading channels," *IEEE Access*, vol. 6, pp. 3068–3084, Feb. 2018.

- Kyusung Shim, **Tri Nhu Do**, Beongku An, "Improving Physical Layer Security of NOMA Networks by Using Opportunistic Scheduling," in *Proc. 2018 International Conference on Ubiquitous and Future Networks (ICUFN)*, Prague, Czech Republic, Jul. 2018, pp. 262–267. —**Best Paper Award**

- Kyusung Shim, Hyukchun Oh, **Tri Nhu Do**, Beongku An, "A Physical Layer Security-Based Transmit Antenna Selection Scheme for NOMA Systems," in *Proc. 2018 International Conference on Ubiquitous and Future Networks (ICUFN)*, Prague, Czech Republic, Jul. 2018, pp. 597–602.

- **Tri Nhu Do**, and Beongku An, "Optimal sum-throughput analysis for downlink cooperative SWIPT NOMA systems," in *Proc. 2018 IEEE International Conference on Recent Advances in Signal Processing, Telecommunications & Computing (SigTelCom)*, Ho Chi Minh city, Vietnam, Jan. 2018, pp. 85–90.

- Kyusung Shim, **Tri Nhu Do**, Beongku An, and Sang-Yep Nam, "A Wireless Power Transfer-based Routing Protocol for Heterogeneous Wireless Networks," in *Proc. 2018 International Conference on Green and Human Information Technology (ICGHIT)*, Chiang Mai, Thailand, Feb. 2018, pp. 75–79. —**Best Paper Award**

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "A physical layer security-based routing protocol in mobile ad-hoc wireless networks," in *Proc. 20th IEEE International Conference on Advanced Communications Technology (ICACT)*, Gangwon-do, South Korea, Feb. 2018, pp. 417–422. —**Outstanding Paper Award**

2017

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, Vo Nguyen Quoc Bao, and Beongku An, "Exploiting direct links in multiuser multirelay SWIPT cooperative networks with opportunistic scheduling," *IEEE Transactions on Wireless Communications*, vol. 16, no. 8, pp. 5410–5427, Aug. 2017.

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "A BNBF user selection scheme for NOMA-based cooperative relaying systems with SWIPT," *IEEE Communications Letters*, vol. 21, no. 3, pp. 664–667, Mar. 2017.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Performance analysis of physical layer security of opportunistic scheduling in multiuser multirelay cooperative networks," *Sensors*, vol. 17, no. 2, pp. 377–396, Feb. 2017.

- **Tri Nhu Do** and Beongku An, "Finding optimal power splitting ratio for cooperative SWIPT NOMA systems," in *Proc. KSII The 9th International Conference on Internet (ICONI)*, Vientiane, Laos, Dec. 2017, pp. 135–136.

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "A full-duplex cooperative scheme with distributed switch-and-stay combining for NOMA networks," in *Proc. 2017 IEEE Global Communications Conference (GLOBECOM)*, Singapore, Dec. 2017, pp. 1–6.

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, and Beongku An, "Transmit antenna selection schemes for MISO-NOMA cooperative downlink transmissions with hybrid SWIPT protocol," in *Proc. 2017 IEEE International Conference on Communications (ICC)*, Paris, France, May 2017, pp. 1–6.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Secure transmission under considering physical layer secure channel state information in mobile ad-hoc wireless," in *Proc. 2017 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Busan, South Korea, Jun. 2017, pp. 1282–1283.

- **Tri Nhu Do**, Daniel Benevides da Costa, Trung Q. Duong, Vo Nguyen Quoc Bao, and Beongku An, "Opportunistic scheduling for fixed-gain amplify-and-forward-based multiuser multirelay SWIPT cooperative networks," in *Proc. 2017 IEEE International Conference on Recent Advances in Signal Processing, Telecommunications & Computing (SigTelCom)*, Da Nang, Jan. 2017, pp. 49–54.

2016

- **Tri Nhu Do**, Daniel Benevides da Costa, and Beongku An, "Performance analysis of multirelay RF energy harvesting cooperative networks with hardware impairments," *IET Communications*, vol. 10, no. 18, pp. 2551–2558, Jun. 2016.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "The impact of hardware impairments and imperfect channel state information on physical layer security," *Journal of the Institute of Electronics and Information Engineers*, vol. 53, no. 4, p. 79–86, Apr. 2016.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "An analysis modeling architecture for supporting physical layer security of wireless networks under hardware impairments," *International Journal of Security and Its Applications*, vol. 10, no. 6, p. 351–362, 2016.

- **Tri Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "Outage performance analysis of relay selection schemes in wireless energy harvesting cooperative networks over non-identical rayleigh fading channels," *Sensors*, vol. 16, no. 3, pp. 295–314, Feb. 2016.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "A source relay selection scheme for physical layer security in multi-user multi-relay amplify-and-forward cooperative networks," in *Proc. 2017 International Conference on Green and Human Information Technology (ICGHIT)*, Hangzhou, China, Feb. 2016, pp. 339–340.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Physical layer security under hard impairments in cooperative communication environments," in *Proc. 2016 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2016, pp. 1641–1643.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Optimal joint source and relay selection scheme for physical layer security in multi-user and multi-relay networks," in *Proc. KSII The 8th International Conference on Internet (ICONI)*, Hong Kong, Jul. 2016, pp. 187–188.

- Le The Dung, **Tri Nhu Do**, and Beongku An, "Cooperative spectrum sensing based routing protocol in mobile cognitive radio ad-hoc networks," in *Proc. 2016 International Conference on Green and Human Information Technology (ICGHIT)*, Angeles City, Philippines, Feb. 2016, pp. 124–125.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Physical layer security of wireless networks under hardware impairments," in *Proc. 2016 International Conference on Green and Human Information Technology (ICGHIT)*, Angeles City, Philippines, Feb. 2016, pp. 130–131.

- **Tri Nhu Do**, Le The Dung, Beongku An, and Sang-Yep Nam, "Connectivity of hybrid overlay/underlay cognitive radio ad hoc networks," in *Proc. 2016 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Da Nang, Vietnam, Jan. 2016, pp. 1–4.

- K. Shim, **Tri Nhu Do**, Beongku An, and Sang-Yep Nam, "Outage performance of physical layer security for multi-hop underlay cognitive radio networks with imperfect channel state information," in *Proc. 2016 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Da Nang, Vietnam, Jan. 2016, pp. 1–4.

2015

- **Tri Nhu Do**, and Beongku An, "A soft-hard combination-based cooperative spectrum sensing scheme for cognitive radio networks," *Sensors*, vol. 15, no. 2, pp. 4388–4407, Feb. 2015.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Physical layer security under hardware impairment and imperfect CSI in relaying networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Gwangju, South Korea, Nov. 2015, pp. 667-668.

- **Tri Nhu Do** and Beongku An, "Outage analysis of best relay selection scheme for wireless energy harvesting relay networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Gwangju, South Korea, Nov. 2015, pp. 669–670.

- **Tri Nhu Do**, Vo Nguyen Quoc Bao, and Beongku An, "A relay selection protocol for wireless energy harvesting relay networks," in *Proc. 2015 IEEE International Conference on Advanced Technologies for Communications*, Ho Chi Minh city, Vietnam, Oct. 2015, pp. 243–247.

- Kyusung Shim, **Tri Nhu Do**, and Beongku An, "Secure transmission using DF on underlay cognitive radio," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2015, pp. 1418–1420.

- **Tri Nhu Do** and Beongku An, "A physical layer security-based multihop relaying protocol for underlay cognitive radio networks," in *Proc. 2015 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2015, pp. 1454–1455.

- **Tri Nhu Do** and Beongku An, "Secure transmission using decode-and-forward protocol for underlay cognitive radio networks," in *Proc. 2015 IEEE International Conference on Ubiquitous and Future Networks (ICUFN)*, Sapporo, Japan, Jul. 2015, pp. 914–918.

- **Tri Nhu Do**, Beongku An, and Sang-Yep Nam, "A likelihood ratio test based hard combination scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2015 International Conference on Green and Human Information Technology (ICGHIT)*, Da Nang, Vietnam, Feb. 2015, pp. 195–196.

- **Tri Nhu Do** and Beongku An, "A comparison of Neyman-Pearson and minimum sensing error criterions for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2015 International Conference on Green and Human Information Technology (ICGHIT)*, Da Nang, Vietnam, Feb. 2015, pp. 181–182.

- **Tri Nhu Do** and Beongku An, "Hybrid cooperative spectrum sensing scheme for cognitive radio networks," in *Proc. 2015 IEEE International Conference on Information Networking (ICOIN)*, Siem Reap, Cambodia, Jan. 2015, pp. 390–391.

2014

- **Tri Nhu Do**, and Beongku An, "Cooperative spectrum sensing schemes with the interference constraint in cognitive radio networks," *Sensors*, vol. 14, no. 5, pp. 8037–8056, May 2014.
- **Tri Nhu Do** and Beongku An, "A location information-based data fusion scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Seoul, South Korea, Nov. 2014, pp. 726–727.
- **Tri Nhu Do** and Beongku An, "Log-likelihood ratio test based spectrum sensing for cognitive radio in small-scale fading channels," in *Proc. 2014 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2014, pp. 1472–1473.
- **Tri Nhu Do** and Beongku An, "Cooperative spectrum sensing schemes with the interference constraint in cognitive radio networks," in *Proc. 2014 IEEE International Symposium on Computer, Consumer and Control (IS3C)*, Taichung, Taiwan, Jun. 2014, pp. 1010–1013.
- **Tri Nhu Do** and Beongku An, "Location-based user selection scheme for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 International Conference on Green and Human Information Technology (ICGHIT)*, Ho Chi Minh City, Vietnam, Feb. 2014, pp. 355–356.
- **Tri Nhu Do** and Beongku An, "Underlay approach-based soft combination schemes for cooperative spectrum sensing in cognitive radio networks," in *Proc. 2014 International Conference on Green and Human Information Technology (ICGHIT)*, Ho Chi Minh City, Vietnam, Feb. 2014, pp. 201–202.
- **Tri Nhu Do**, Beongku An, and Kyu-Tae Lee, "A cooperative spectrum sensing scheme using the concept of underlay cognitive radio," in *Proc. 2014 IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Kota Kinabalu, Malaysia, Jan. 2014, pp. 1–3.

2013

- **Tri Nhu Do** and Beongku An, "Cooperative spectrum sensing in cognitive radio networks under interference constraint," in *Proc. 2013 The Institute of Electronics Engineers of Korea (IEEK) Fall Conference*, Seoul, South Korea, Nov. 2013, pp. 902–903.
- **Tri Nhu Do** and Beongku An, "A discussion of cooperative spectrum sensing using Alamouti code in cognitive radio networks," in *Proc. 2013 The Institute of Electronics Engineers of Korea (IEEK) Summer Conference*, Jeju Island, South Korea, Jun. 2013, pp. 1437–1438.

Honors and Awards

- **Top reviewers in Computer Science**, awarded by Publons in Sep. 2018 and Sep. 2019.
- **Best Paper Award** in the 10th IEEE International Conference on Ubiquitous and Future Networks (ICUFN), Prague, Czech Republic, Jul. 2018
- **Best Paper Award** in the 6th International Conference on Green and Human Information Technology (ICGHIT), Chiang Mai, Thailand, Feb. 2018
- **Outstanding Paper Award** in the 20th IEEE International Conference on Advanced Communications Technology (ICACT), Gangwon-do, South Korea, Feb. 2018
- **Global Scholarship** awarded from Hongik University, South Korea Mar. 2015 – Dec. 2016
- **Global Scholarship** awarded from Hongik University, South Korea Mar. 2013 – Dec. 2014

Services

Technical Program Committee (TPC) Member

- 2020 IEEE Vehicular Technology Conference: VTC2020-Spring
- 2019 IEEE Global Communications Conference
- 2018 IEEE Vehicular Technology Conference: VTC2018-Fall
- 2018 IEEE International Conference on Advanced Networks and Telecommunications Systems

Reviewer

I have been serving as reviewer for top-tier journals and conferences as recored at *Review Record–Publons.com (link)*

References

Prof. Zygmunt J. Haas
(*Postdoctoral supervisor*)
Professor & Distinguished Chair in Computer Science
Erik Jonsson School of Engineering and Computer Science
University of Texas at Dallas
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Prof. Beongku An
(*PhD advisor*)
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Hongik University
2639, Sejong-ro, Jochiwon-eup
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Prof. Daniel B. da Costa
(*Collaborative researcher*)
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